A Sharp Capacity Criterion for Heterogeneous Four-State Potts Broadcasting on Trees

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Abstract

For the ferromagnetic four-state Potts broadcast process on an arbitrary locally finite deterministic rooted tree, suppose the edge-dependent second eigenvalues satisfy $0 < \lambda_e \le q < 1$. We show that reconstruction occurs if and only if the associated L^3 nonlinear capacity is positive, with path weights given by the products of squared edge eigenvalues. Neither a uniform vertex-degree bound nor a positive lower bound on the channel parameters is required. We establish sharp comparability, up to constants depending only on the uniform contraction bound, between quadratic root information and capacity; total-variation advantage then follows by the optimal elementary conversions. Robust reconstruction is equivalent to the same capacity condition. We also derive an exact series criterion for spherically symmetric trees, including at the Kesten–Stigum critical boundary. The proof rests on the recent four-state posterior moment-saving inequality. A cubic correction converts that input into a degree-free nonlinear flow recursion. Conversely, an elementary two-moment calculation, a controlled independent erasure, and an exact flow-averaged path identity give reconstruction from positive capacity without invoking the imported inequality. Beyond the uniformly contractive regime we prove a necessary criterion in terms of an edge-noise-weighted capacity, which is exact on one ray; the full weighted-capacity converse remains open.

Keywords. Broadcasting on trees; four-state Potts model; reconstruction; nonlinear capacity; heterogeneous channels; information flow.

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1 Introduction

Reconstruction on trees asks whether the spin at the root can be inferred with a nonvanishing advantage from the spins on arbitrarily distant levels. The Kesten–Stigum second-eigenvalue scale originates in multitype branching-process theory [1]; broadcasting and information flow on trees were developed in [2, 3]. For the binary Ising model, nonlinear potential and capacity furnish finer criteria on nonregular trees, including at criticality. The general-tree capacity theory of Pemantle and Peres [6] is an important antecedent: in its Ising setting, an L^3 capacity occurs for plus boundary conditions. Robust reconstruction and the role of the second eigenvalue were investigated by Janson and Mossel [4].

For the symmetric Potts broadcast channel, the transition law is a convex combination of the identity and the uniform kernel. The four-state case lies at a cancellation point in the posterior expansion; the standard quadratic correction is insufficient and a higher-order term controls the critical regime. The sharpness and nonsharpness phenomena across the number of states were studied by Sly [7], with later results of Mossel, Sly and Sohn [8]. Two recent manuscripts [9, 10] provide, respectively, a rigorously certified four-state posterior moment inequality and an L^3 criterion for bounded-degree,

constant-channel deterministic trees. The present note extends the latter to heterogeneous channels and removes the bounded-degree restriction, while preserving its explicit dependence on the former.

Our two principal transfers are different. The *necessity* argument modifies the information variable by $f(m) = m + m^3/3000$. The moment-saving inequality makes this variable subadditive under every binary combination, so sequential products can be controlled with constants independent of the number of children. The resulting edge-dependent concave recursion creates an admissible flow by telescoping squared increments, in the spirit of nonlinear capacity recursions [6].

In the *sufficiency* argument we calibrate deliberately noisy boundary observations to an arbitrary admissible flow. Unlike the necessity proof, this requires only elementary identities for the quadratic information and a signed cubic moment of symmetric posterior laws. Independent erasure ensures nonnegative information loss without the imported cone inequality. Sampling a ray from the flow converts the total loss into a path sum, and the exact propagation factors leave a kernel bounded by q^{distance}. This gives a uniform lower bound on root information independently of the depth and of vertex degrees. Deliberately degraded boundary observations also appear in robust reconstruction [4]; the present construction calibrates their strengths separately at every vertex using nonlinear capacity. Because its terminal degradation may be placed after any fixed additional leaf noise, the same argument proves robust reconstruction.

2 Model, capacity, and main theorem

Let T be an infinite locally finite rooted tree with root o and levels $L_n = \{v : |v| = n\}$. Assign every edge e a parameter

$$0 < \lambda_e \le q < 1. \tag{2.1}$$

The tree and the parameters are deterministic and known to the observer. Start with a uniform root spin in [4] and, independently along edges conditional on the parent spins, use the transition matrix

$$P_e(j \mid i) = \lambda_e \mathbf{1}_{\{j=i\}} + \frac{1-\lambda_e}{4}.$$

Write $\mathsf{u} = (1/4, 1/4, 1/4, 1/4)$, let $p_o^{(n)}$ be the posterior vector of the root given σ_{L_n}, and put

$$a_n = \mathbb{E}\ \mathrm{TV}(p_o^{(n)}, \mathsf{u}), \qquad m_n = 4\mathbb{E}\|p_o^{(n)} - \mathsf{u}\|_2^2.$$

Both sequences are nonincreasing by data processing; write their limits as a_∞ and m_∞. Reconstruction means $a_\infty > 0$, equivalently $m_\infty > 0$.

Put

$$\rho(o) = 1, \qquad \rho(v) = \prod_{e \in [o,v]} \lambda_e^2. \tag{2.2}$$

A nonnegative flow θ to infinity is conserved at every nonroot vertex, including vertices with no children. Its mass is $|\theta| = \sum_{w \in \mathrm{ch}(o)} \theta(o, w)$. For $v \ne o$, write e_v for its incoming edge. Define

$$\mathrm{Cap}_3(T, \rho) = \sup \left\{ |\theta| : \sup_{\xi \in \partial T} \sum_{v \in \xi \setminus \{o\}} \left(\frac{\theta(e_v)}{\rho(v)} \right)^2 \le 1 \right\}. \tag{2.3}$$

The zero flow is allowed. This is the standard L^3 convention in which the path-potential exponent is $3 - 1 = 2$.

Theorem 2.1. *Under (2.1), for every infinite locally finite deterministic rooted tree,*

$$\boxed{a_\infty > 0 \quad \Longleftrightarrow \quad m_\infty > 0 \quad \Longleftrightarrow \quad \mathrm{Cap}_3(T,\rho) > 0.}$$

Put

$$\delta_q = \frac{(1-q^4)/3000}{(1+9/3000)^3}.$$

Then the quadratic information satisfies the two-sided comparison

$$\frac{1-q}{2512}\min\{\mathrm{Cap}_3(T,\rho),1\} \le m_\infty \le \frac{1}{\sqrt{2\delta_q}}\min\{\mathrm{Cap}_3(T,\rho),1\}. \tag{2.4}$$

Consequently,

$$\frac{1-q}{20096}\min\{\mathrm{Cap}_3(T,\rho),1\} \le a_\infty \le \frac{1}{2(2\delta_q)^{1/4}}\sqrt{\min\{\mathrm{Cap}_3(T,\rho),1\}}. \tag{2.5}$$

For $0 < \eta \le 1$, let $a_n^{(\eta)}$ denote the reconstruction advantage when every level-n spin is first passed independently through an additional four-state Potts channel of parameter η. Write $a_\infty^{(\eta)} = \lim_n a_n^{(\eta)}$.

Corollary 2.2 (Robust reconstruction)**.** *Under (2.1), for every fixed $0 < \eta \le 1$,*

$$a_\infty^{(\eta)} > 0 \quad \Longleftrightarrow \quad \mathrm{Cap}_3(T,\rho) > 0.$$

More precisely,

$$a_\infty^{(\eta)} \ge \frac{1}{32}\min\left\{\frac{1-q}{628},\, 3\eta^2\right\}\min\{\mathrm{Cap}_3(T,\rho),1\}. \tag{2.6}$$

The theorem in [10] assumes a common channel parameter and a uniform degree bound. The present proof retains its information-flow strategy but strengthens both local steps. For necessity, the only imported probabilistic result is Proposition 3.2 of [9], stated precisely in Proposition 3.1 below from the source version pinned in the bibliography. Its source proof uses exact polynomial certificates. We do not claim to replace that certificate proof. Everything after the stated input is proved here. In particular, the capacity-to-reconstruction direction is independent of that input.

3 Posterior coordinates and two local inequalities

An experiment observes a random variable about a uniform input spin. Its posterior law is the law of its posterior vector p under the marginal observation distribution. It is symmetric if it is invariant under permutations of the four spin labels. Set

$$L = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad x = Lp,$$

and define

$$m(\mu) = \mathbb{E}_\mu|x|^2, \qquad \tau(\mu) = \mathbb{E}_\mu x_1x_2x_3, \qquad h(\mu) = |\tau(\mu)|. \tag{3.1}$$

The following bounds will be used repeatedly:

$$|x|^2 = 4\|p-\mathsf{u}\|_2^2, \quad 0 \le m \le 3, \quad h \le \frac{m}{3}, \qquad \frac{m}{8} \le \mathbb{E}\,\mathrm{TV}(p,\mathsf{u}) \le \frac{1}{2}\sqrt{m}. \tag{3.2}$$

The cubic bound follows pointwise from $3|x_1x_2x_3| \le x_1^2 + x_2^2 + x_3^2$, by AM–GM and $|x_i| \le 1$. The lower total-variation bound follows from $\sum_i (p_i - 1/4)^2 \le \sum_i |p_i - 1/4|$, while the upper bound follows from Cauchy–Schwarz and Jensen. If an observation is degraded, its posterior becomes a conditional expectation of the old posterior, so m cannot increase.

A channel with parameter $t \in [0,1]$ changes the posterior to $\mathsf{u} + t(p - \mathsf{u})$ and hence

$$x \mapsto tx, \qquad m \mapsto t^2 m, \qquad \tau \mapsto t^3 \tau, \qquad h \mapsto t^3 h. \tag{3.3}$$

The revealing experiment has $m = 3$ and $\tau = 1$.

Proposition 3.1 (Imported moment-saving input)**.** *There is a class $\mathcal{C}$ of symmetric posterior laws containing the revealing and uninformative laws and closed under every channel step $t \in [0,1]$ and under conditionally independent products. If $\mu, \nu \in \mathcal{C}$, write $u = m(\mu)$ and $v = m(\nu)$, and let $x = Lp$ and $y = Lr$ under their respective posterior laws. Then*

$$m(\mu \star \nu) \le u + v - \frac{\mathbb{E}_\mu |x|^4 \, v + u \, \mathbb{E}_\nu |y|^4}{1000}. \tag{3.4}$$

Consequently, with $c = 1/1000$,

$$m(\mu \star \nu) \le u + v - cuv(u + v). \tag{3.5}$$

The first assertion is Proposition 3.2 and its closure statement in [9]. It explicitly permits arbitrary channel parameters and nonidentical input laws. The consequence (3.5) follows from Jensen's inequalities $\mathbb{E}|x|^4 \ge u^2$ and $\mathbb{E}|y|^4 \ge v^2$. Every finite subtree broadcast experiment and every partial product of its branches belongs to $\mathcal{C}$. Limiting posterior laws will not need to be in $\mathcal{C}$.

The next lemma, unlike Proposition 3.1, needs only symmetry. It strengthens the smoothing-dependent estimate in [10, Lemma 4.1].

Lemma 3.2 (Universal local moment bounds)**.** *For any two symmetric posterior laws, put $u = m(\mu)$ and $v = m(\nu)$. Then*

$$m(\mu \star \nu) \ge u + v - 5\big(h(\mu)v + uh(\nu)\big), \tag{3.6}$$

$$h(\mu \star \nu) \le h(\mu) + h(\nu) + 376uv. \tag{3.7}$$

No smoothing or membership in $\mathcal{C}$ is assumed.

Proof. Take independent marginal posterior vectors p, r, and write $x = Lp$, $y = Lr$, and $s = x \cdot y$. The joint marginal distribution of the observations is their independent marginal product weighted by (cf. the posterior-product formalism in [5])

$$Z = 4 \sum_i p_i r_i = 1 + s.$$

When $Z > 0$, the combined posterior has coordinates

$$LP = \frac{N}{Z}, \qquad N = x + y + R, \qquad R_i = x_j y_k + x_k y_j, \quad \{i,j,k\} = \{1,2,3\}. \tag{3.8}$$

In particular, $|N_i| \le Z$ and $0 \le Z \le 4$. At $Z = 0$, all N_i vanish; all weighted posterior quantities below are assigned zero. Their bounds show that these extensions are continuous.

Symmetry acts on x by coordinate permutations and even sign changes. Consequently

$$\mathbb{E} x_i = 0, \qquad \mathbb{E} x_i x_j = \frac{u}{3} \mathbf{1}_{\{i=j\}},$$

and among degree-three monomials only $x_1x_2x_3$ can have nonzero expectation. Put $\zeta_\mu = \tau(\mu)$ and $\zeta_\nu = \tau(\nu)$. Direct expansion gives

$$\mathbb{E}|R|^2 = \frac{2}{3}uv, \qquad \mathbb{E}s^2 = \frac{1}{3}uv,$$

$$\mathbb{E}[s\,x\cdot R] = 2\zeta_\mu v, \qquad \mathbb{E}[s\,y\cdot R] = 2u\zeta_\nu, \qquad \mathbb{E}[s|R|^2] = 6\zeta_\mu\zeta_\nu.$$

Using $Z^{-1} = 1 - s + s^2/Z$, with the marginal weight Z included, we obtain the exact identity

$$m(\mu\star\nu) = u + v - 4(\zeta_\mu v + u\zeta_\nu) - 6\zeta_\mu\zeta_\nu + \mathbb{E}\frac{|N|^2 s^2}{Z}. \tag{3.9}$$

The mixed quadratic terms cancel since $\mathbb{E}|R|^2 = 2\mathbb{E}s^2$, and the remainder is nonnegative. By $h(\mu) \le u/3$ and $h(\nu) \le v/3$,

$$6|\zeta_\mu\zeta_\nu| \le h(\mu)v + uh(\nu).$$

Equation (3.6) follows.

For the third moment,

$$\tau(\mu\star\nu) = \mathbb{E}\frac{N_1N_2N_3}{Z^2}, \qquad Z^{-2} - (1-2s) = \frac{s^2(3+2s)}{Z^2}.$$

Because $|N_i| \le Z$, $-1 \le s \le 3$, and $Z \le 4$, the absolute remainder after replacing Z^{-2} by $1-2s$ is at most

$$36\mathbb{E}s^2 = 12uv.$$

An exact expansion of $N_1N_2N_3(1-2s)$ has two pure terms, $x_1x_2x_3$ and $y_1y_2y_3$, each with coefficient one. Mixed terms of degree one in either vector average to zero. The sum of the absolute coefficients of all remaining mixed terms is 364. Every such term has degree at least two in each vector and absolute expectation at most uv: all coordinate absolute values are at most one, and retaining two factors bounds each marginal monomial by the corresponding squared norm. Hence the polynomial expectation differs from $\zeta_\mu + \zeta_\nu$ by at most $364uv$. Adding the remainder proves (3.7). The coefficient count is an exact integer expansion and is reproduced by the verification script accompanying the source.

All expressions at $Z = 0$ extend as asserted because they are bounded by a constant times Z, possibly multiplied by a bounded power of s. □

4 Reconstruction implies capacity

The necessity argument has a useful form that does not require a uniform contraction bound. For arbitrary parameters $0 < \lambda_e \le 1$, retain (2.2), put

$$\omega_e = 1 - \lambda_e^4,$$

and define the edge-noise-weighted capacity

$$\mathrm{Cap}_3^{\mathrm{noise}}(T,\rho) = \sup\left\{|\theta| : \sup_{\xi\in\partial T}\sum_{v\in\xi\setminus\{o\}}\omega_{e_v}\left(\frac{\theta(e_v)}{\rho(v)}\right)^2 \le 1\right\}. \tag{4.1}$$

4.1 A cubic statistic removes degree dependence

Set

$$K = \frac{c}{3} = \frac{1}{3000}, \qquad f(t) = t + Kt^3, \qquad D_f = 1 + 9K. \tag{4.2}$$

If $t = m(\mu \star \nu)$ for laws in $\mathcal{C}$, then (3.5) and $t \le u + v$ give

$$\begin{aligned} f(t) &\le u + v - cuv(u+v) + K(u+v)^3 \\ &= f(u) + f(v) + (3K - c)uv(u+v) = f(u) + f(v). \end{aligned}$$

Thus $f(m)$ is subadditive under any finite product. The number of children never enters the constant.

Let $m_v^{(n)}$ be the information at v from descendants at distance n, and set $m_v = \lim_n m_v^{(n)}$. Data processing gives these limits in $[0, 3]$. For $r_{vw} = \lambda_{vw}^2$, channel scaling, subadditivity of $f(m)$, and passage to the limit imply

$$f(m_v) \le \sum_{w \in \mathrm{ch}(v)} f(r_{vw} m_w). \tag{4.3}$$

All sums are finite. No closure assertion about a limiting posterior is used.

Put $y_v = f(m_v)$, set $\omega_{vw} = 1 - \lambda_{vw}^4$, and define

$$\delta_{vw} = \frac{K(1 - r_{vw}^2)}{D_f^3} = \frac{K\omega_{vw}}{D_f^3}.$$

Since $y_w \le D_f m_w$ and

$$f(rm) = rf(m) - Kr(1 - r^2)m^3,$$

equation (4.3) yields

$$y_v \le \sum_w r_{vw} y_w (1 - \delta_{vw} y_w^2) \le \sum_w \frac{r_{vw} y_w}{\sqrt{1 + 2\delta_{vw} y_w^2}}. \tag{4.4}$$

The last inequality uses $1 - t \le (1 + 2t)^{-1/2}$ for $t \ge 0$.

4.2 The edge-weighted flow

Choose coefficients

$$0 \le d_{vw} \le \frac{r_{vw}}{\sqrt{1 + 2\delta_{vw} y_w^2}}, \qquad y_v = \sum_w d_{vw} y_w.$$

They are obtained by multiplying all upper-bound coefficients by the ratio of y_v to the right side of (4.4); if that side is zero, use zero coefficients. Define

$$\ell_o = 1, \qquad \ell_w = \ell_v d_{vw}, \qquad \vartheta(v, w) = \ell_w y_w.$$

These equations give flow conservation and root mass $y_o = f(m_\infty)$. With $g_v = \ell_v / \rho(v)$ and $\rho(w) = \rho(v) r_{vw}$,

$$g_w^2 (1 + 2\delta_{vw} y_w^2) \le g_v^2, \qquad \frac{\vartheta(v, w)}{\rho(w)} = g_w y_w.$$

Therefore

$$\omega_{vw} \left(\frac{\vartheta(v, w)}{\rho(w)} \right)^2 \le \frac{D_f^3}{2K} (g_v^2 - g_w^2).$$

Summing along a ray telescopes. After scaling the flow, we obtain the following quantitative obstruction.

Theorem 4.1 (Weighted-capacity necessity)**.** *For an arbitrary locally finite deterministic rooted tree with $0 < \lambda_e \le 1$, reconstruction implies*

$$\mathrm{Cap}_3^{\mathrm{noise}}(T,\rho) \ge \sqrt{\frac{2K}{D_f^3}}\, f(m_\infty) \ge \sqrt{\frac{2K}{D_f^3}}\, m_\infty > 0. \tag{4.5}$$

Under the uniform bound (2.1), one has $\omega_e \ge 1 - q^4$. Hence a weighted-admissible flow scaled by $\sqrt{1-q^4}$ is admissible for (2.3). Consequently,

$$\mathrm{Cap}_3(T,\rho) \ge \sqrt{2\delta_q}\, f(m_\infty) \ge \sqrt{2\delta_q}\, m_\infty. \tag{4.6}$$

This proves the necessity implication in Theorem 2.1. It also gives $m_\infty \le \mathrm{Cap}_3(T,\rho)/\sqrt{2\delta_q}$. Since $m_\infty \le 3$ and $(2\delta_q)^{-1/2} > 3$, this is the upper bound in (2.4). The upper total-variation bound in (2.5) follows from $a_\infty \le \frac{1}{2}\sqrt{m_\infty}$.

5 Positive capacity implies reconstruction

This direction uses only Lemma 3.2. We introduce a harmless optional erasure to ensure nonnegative information deficits without assuming Proposition 3.1.

5.1 Products followed by erasure

For symmetric experiments with information u, v, form their raw product, whose information is m_*. If $m_* > u + v$, retain the combined observation with probability $(u+v)/m_*$ and otherwise output a fixed erasure symbol; if $m_* \le u + v$, retain it always. The retention coin is independent of the spin and of all observations; its parameter depends only on the experiment laws. Conditional on erasure, the posterior is uniform. Thus the new experiment has information

$$m' = \min\{m_*, u + v\}$$

and third-moment magnitude at most that of the raw product. Lemma 3.2 gives

$$0 \le u + v - m' \le 5(h_1 v + u h_2), \qquad h' \le h_1 + h_2 + 376uv. \tag{5.1}$$

The experiment remains symmetric and is a degradation of the original pair. Independent random coins in disjoint subtrees preserve the conditional independence needed at subsequent products.

5.2 A flow profile and finite observations

Take an admissible flow of mass $\alpha > 0$, and scale it down so that $\alpha \le 1$. For $0 < \varepsilon \le 1$, set

$$\Theta(v) = \varepsilon\theta(e_v)\ (v \ne o), \qquad \Theta(o) = M = \varepsilon\alpha, \qquad X_v = \frac{\Theta(v)}{\rho(v)}.$$

Flow conservation says

$$X_v = \sum_{w \in \mathrm{ch}(v)} \lambda_{vw}^2 X_w. \tag{5.2}$$

Every positive-flow edge extends to an infinite ray, so admissibility gives $X_v \le \varepsilon$ for $v \ne o$. Also $X_o = M \le \varepsilon$. On every ray,

$$\sum_{v \in \xi} X_v^2 \le 2\varepsilon^2. \tag{5.3}$$

This normalization replaces the uniform degree bound used to control the root in the homogeneous argument.

Fix a depth n. At each $v \in L_n$, pass its spin through an additional independent channel of parameter $t_v = \sqrt{X_v/3}$. The terminal experiment then has

$$\widehat{m}_v = X_v, \qquad \widehat{h}_v = (X_v/3)^{3/2} \le X_v/3. \tag{5.4}$$

Proceed upward. At each vertex pass each constructed child experiment through its actual edge channel, and combine the branches sequentially using (5.1). All experiments so obtained are degradations of the level-n observation. Their information satisfies

$$0 \le \widehat{m}_v \le X_v \tag{5.5}$$

by induction and (5.2). Empty observations and zero-flow subtrees cause no difficulty.

5.3 Local deficits with constants independent of degree

At an internal vertex, write

$$b_w = \lambda_{vw}^2 \widehat{m}_w, \qquad h'_w = \lambda_{vw}^3 \widehat{h}_w, \qquad S = \sum_w b_w \le X_v, \qquad H = \sum_w h'_w.$$

Let

$$D_v = \sum_w \lambda_{vw}^2 \widehat{m}_w - \widehat{m}_v. \tag{5.6}$$

With $C_0 = 376$, sequential use of (5.1) gives

$$\widehat{h}_v \le H + \frac{C_0}{2} S^2, \qquad 0 \le D_v \le 5SH + \frac{5C_0}{6} S^3. \tag{5.7}$$

Here are the details of the degree-independent sums. Order the children and put $s_{j-1} = \sum_{i<j} b_i$, $H_{j-1} = \sum_{i<j} h'_i$. The partial product has $m \le s_{j-1}$ and $h \le H_{j-1} + (C_0/2) s_{j-1}^2$. The accumulated third-moment errors are bounded by $C_0 \sum_j s_{j-1} b_j \le (C_0/2) S^2$. The mixed loss terms satisfy

$$\sum_j (H_{j-1} b_j + s_{j-1} h'_j) \le SH, \qquad \sum_j s_{j-1}^2 b_j \le \int_0^S t^2 \, dt = \frac{S^3}{3}.$$

These are exactly (5.7).

Fix $A = 314$. Since $A \ge C_0/2$, $A \ge 5C_0/6$, and $A \ge 5$, we may use the simpler bounds

$$\widehat{h}_v \le \sum_w \lambda_{vw}^3 \widehat{h}_w + A X_v^2, \tag{5.8}$$

$$0 \le D_v \le A X_v \left(\sum_w \lambda_{vw}^3 \widehat{h}_w + X_v^2 \right). \tag{5.9}$$

Crucially, the propagated third moments have their exact channel factors; the constant A multiplies only newly inserted errors.

For a descendant u of v, put

$$\Lambda(v, u) = \prod_{e \in [v,u]} \lambda_e, \qquad \Lambda(v, v) = 1.$$

Unfolding (5.8) in (5.9), and using (5.4), gives

$$D_v \le A^2 X_v \sum_{\substack{u \succeq v \\ |u|<n}} \Lambda(v,u)^3 X_u^2 + \frac{A}{3} X_v \sum_{\substack{u \succeq v \\ |u|=n}} \Lambda(v,u)^3 X_u. \tag{5.10}$$

There is no repeated multiplication by A along the depth of the tree. Each error is inserted once and thereafter propagated only by channel products.

Multiplying (5.6) by $\rho(v)$ and summing over internal vertices telescopes, because $\rho(w) = \rho(v)\lambda_{vw}^2$:

$$\widehat{m}_o = M - \sum_{|v|<n} \rho(v) D_v. \tag{5.11}$$

5.4 The heterogeneous path identity

Choose $V_n \in L_n$ with probability $\Theta(V_n)/M$, and let $V_0 = o, V_1, \dots, V_n$ be its ancestors. Conservation gives

$$\mathbb{P}(V_j = u) = \frac{\Theta(u)}{M}. \tag{5.12}$$

Every path with positive probability extends to an infinite ray and satisfies (5.3). The exact identity

$$\boxed{\rho(v)\Lambda(v,u)^3 X_u = \Lambda(v,u)\Theta(u)} \tag{5.13}$$

is the required transfer of the third-moment propagation to the flow. Indeed, $\rho(u) = \rho(v)\Lambda(v,u)^2$.

For fixed depths $i \le j < n$, summing over descendants in the first term of (5.10) becomes

$$\sum_{|v|=i} \rho(v) X_v \sum_{\substack{u \succeq v \\ |u|=j}} \Lambda(v,u)^3 X_u^2 = M\mathbb{E}[\Lambda(V_i,V_j) X_{V_i} X_{V_j}].$$

The terminal term similarly becomes $M\mathbb{E}[\Lambda(V_i,V_n)X_{V_i}]$. Hence

$$\frac{1}{M}\sum_{|v|<n} \rho(v) D_v \le A^2 \mathbb{E} \sum_{0 \le i \le j < n} \Lambda(V_i,V_j) X_{V_i} X_{V_j} + \frac{A}{3}\mathbb{E}\sum_{i<n} \Lambda(V_i,V_n) X_{V_i}. \tag{5.14}$$

The kernel is path-dependent, but $\Lambda(V_i,V_j) \le q^{j-i}$. For each fixed $k \ge 0$, Cauchy–Schwarz and (5.3) imply pathwise

$$\sum_i X_{V_i} X_{V_{i+k}} \le 2\varepsilon^2.$$

Also $X_{V_i} \le \varepsilon$. Therefore (5.14) is at most

$$\frac{2A^2\varepsilon^2}{1-q} + \frac{Aq\varepsilon}{3(1-q)}. \tag{5.15}$$

Choose once and for all

$$\varepsilon = \frac{1-q}{2A} = \frac{1-q}{628}. \tag{5.16}$$

Then (5.15) is at most $(1-q)/2 + q/6 \le 1/2$. In particular, (5.11) gives $\widehat{m}_o \ge M/2$ for every n. The original level observation has at least this information, so

$$a_n \ge \frac{\varepsilon\alpha}{16} > 0 \qquad (n \ge 1). \tag{5.17}$$

This proves reconstruction.

For the quantitative lower bound, write $C = \mathrm{Cap}_3(T, \rho)$. If $C > 0$, an admissible flow can be chosen and scaled to mass $\alpha = \min\{1, C/2\}$, which is at least $\frac{1}{2}\min\{C, 1\}$. (The capacity is finite because the root has finitely many children.) Since the original observation has quadratic information at least $\widehat{m}_o \ge \varepsilon\alpha/2$, we obtain

$$m_\infty \ge \frac{1-q}{2512}\min\{C, 1\}, \qquad a_\infty \ge \frac{1-q}{20096}\min\{C, 1\}.$$

If $C = 0$, the lower bounds are immediate. This completes Theorem 2.1.

Proof of Corollary 2.2. The reverse implication follows from data processing. Assume $C > 0$ and use the same admissible flow. Put

$$\varepsilon_\eta = \min\left\{\frac{1-q}{628},\, 3\eta^2\right\}.$$

Then $X_v \le \varepsilon_\eta \le 3\eta^2$ at every terminal. Starting from the already-noisy terminal spin, whose channel parameter is η, apply the further channel of parameter $\sqrt{X_v/3}/\eta \le 1$. The resulting terminal experiment is exactly the one used in (5.4). All subsequent estimates are unchanged, and the loss bound is no larger than $1/2$ because $\varepsilon_\eta \le (1-q)/628$. Thus

$$a_n^{(\eta)} \ge \frac{\varepsilon_\eta\alpha}{16} \ge \frac{\varepsilon_\eta}{32}\min\{C, 1\}$$

for every n, proving (2.6) and the corollary. □

6 Consequences and the contraction boundary

Remark 6.1 (Quenched random environments). If the rooted tree and its channel parameters are random but, conditional on the environment, form a locally finite deterministic tree satisfying a common bound $\lambda_e \le q < 1$, Theorem 2.1 applies to almost every realization. Thus quenched reconstruction is equivalent realization by realization to positive quenched capacity. This observation does not by itself identify an annealed threshold, because averaging may mix environments with different capacities.

Corollary 6.2 (Spherical trees)**.** *Suppose every vertex at the same depth has the same number of children, let $N_k = |L_k|$, and suppose every depth-k edge has parameter $\lambda_k \le q < 1$. Put $\rho_k = \prod_{j=1}^{k}\lambda_j^2$. Then*

$$\mathrm{Cap}_3(T, \rho) = \left(\sum_{k\ge 1}(N_k\rho_k)^{-2}\right)^{-1/2},$$

where the value is zero if the series diverges. Consequently reconstruction occurs exactly when that series converges.

Proof. The uniform flow of mass a assigns a/N_k to each depth-k edge, giving the displayed path energy. Conversely, choose a uniform ray, so its vertex at level k is uniform on L_k. For any flow of mass a, Jensen gives

$$\mathbb{E}\left(\frac{\theta(e_{V_k})}{\rho_k}\right)^2 \ge \left(\frac{a}{N_k\rho_k}\right)^2.$$

Average the path potential and sum by monotone convergence. Its supremum is at least this average, proving optimality of the uniform flow. □

In particular, if $N_k\rho_k \asymp k^\beta$, the criterion is $\beta > 1/2$. On a regular d-ary tree with common parameter λ, it reduces to $d\lambda^2 > 1$, including nonreconstruction at equality. These are direct consequences of the proved series criterion.

Proposition 6.3 (The weighted capacity is exact on one ray)**.** *Suppose T is one infinite ray, with edge parameters $0 < \lambda_k \le 1$. Put*

$$\Lambda_n = \prod_{k=1}^{n} \lambda_k, \qquad \rho_n = \Lambda_n^2, \qquad \Lambda_\infty = \lim_n \Lambda_n.$$

Then

$$a_\infty = \frac{3}{4}\Lambda_\infty,$$

and

$$\mathrm{Cap}_3^{\mathrm{noise}}(T,\rho) = \begin{cases} (\rho_\infty^{-2} - 1)^{-1/2}, & 0 < \rho_\infty < 1, \\ +\infty, & \rho_\infty = 1, \\ 0, & \rho_\infty = 0. \end{cases} \tag{6.1}$$

Consequently, reconstruction on a ray is equivalent to positive weighted capacity without any uniform contraction assumption.

Proof. Composition of the symmetric channels multiplies their parameters, so the root-to-level-n channel has parameter Λ_n. Observing the terminal spin therefore gives total-variation advantage $3\Lambda_n/4$.

Every flow on a ray is constant, say of mass a. Since $\rho_k = \rho_{k-1}\lambda_k^2$,

$$\frac{1 - \lambda_k^4}{\rho_k^2} = \rho_k^{-2} - \rho_{k-1}^{-2}.$$

Thus its weighted path energy through level n is $a^2(\rho_n^{-2} - 1)$. Letting $n \to \infty$ proves (6.1). □

The uniform upper bound in Theorem 2.1 cannot simply be discarded while retaining the unweighted capacity. For example, on one ray take $\lambda_k = \exp(-2^{-k})$. Then $\Lambda_\infty = e^{-1}$, so $a_\infty = 3/(4e) > 0$. Every nonzero flow has constant mass, whereas each unweighted capacity summand is at least the square of that mass; hence $\mathrm{Cap}_3(T,\rho) = 0$. Proposition 6.3 shows that the edge-noise weight repairs this obstruction exactly on a ray. Theorem 4.1 proves the corresponding necessity on every tree. A general weighted-capacity converse would require a refinement of the sufficiency loss estimate: the geometric-kernel bound used above deliberately collapses long one-child stretches and is not controlled by weighted energy alone.

Remark 6.4 (Zero edges). The statement assumes strictly positive edge parameters so that ρ^{-1} is defined. An edge with $\lambda_e = 0$ separates the spin below it completely from the root. If zero edges are to be admitted, remove their downstream components and use the theorem on the remaining root component; the capacity formula must be interpreted on that component, not by dividing by zero. If the remaining root component is finite, reconstruction is impossible and its capacity is zero.

7 Dependence on computationally certified input

The only imported inequality in the proof of Theorems 2.1 and 4.1 is Proposition 3.1, the preserved-class moment-saving estimate of [9, Proposition 3.2]. The citation is pinned to OpenAI/math commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. That source proves its underlying polynomial inequalities with exact rational coefficient certificates, subdivision, and executable verifiers; see its certificate and finite-verification sections. Consequently, the two necessity arguments are conditional only in the ordinary sense that they invoke this explicitly stated theorem of the cited preprint. The sufficiency implication, robust reconstruction corollary, and their quantitative lower bounds are independent of that imported result.

All remaining estimates, including the universal product identities, the sharpened coefficient count, the cubic subadditivity argument, the weighted flow recursion, and the heterogeneous path summation, are established in the present paper. The exact integer expansion used in Lemma 3.2 is independently checked by `verify_local_expansion.py` in the source directory.

References

[1] H. Kesten and B. P. Stigum, *Additional limit theorems for indecomposable multidimensional Galton–Watson processes*, Ann. Math. Statist. **37** (1966), 1463–1481. doi:10.1214/aoms/1177699139.

[2] W. Evans, C. Kenyon, Y. Peres, and L. J. Schulman, *Broadcasting on trees and the Ising model*, Ann. Appl. Probab. **10** (2000), 410–433. doi:10.1214/aoap/1019487349.

[3] E. Mossel and Y. Peres, *Information flow on trees*, Ann. Appl. Probab. **13** (2003), 817–844. doi:10.1214/aoap/1060202828.

[4] S. Janson and E. Mossel, *Robust reconstruction on trees is determined by the second eigenvalue*, Ann. Probab. **32** (2004), 2630–2649. doi:10.1214/009117904000000153.

[5] M. Mézard and A. Montanari, *Reconstruction on trees and spin glass transition*, J. Stat. Phys. **124** (2006), 1317–1350. doi:10.1007/s10955-006-9162-3.

[6] R. Pemantle and Y. Peres, *The critical Ising model on trees, concave recursions and nonlinear capacity*, Ann. Probab. **38** (2010), 184–206. doi:10.1214/09-AOP482.

[7] A. Sly, *Reconstruction for the Potts model*, Ann. Probab. **39** (2011), 1365–1406. doi:10.1214/10-AOP584.

[8] E. Mossel, A. Sly, and Y. Sohn, *Exact phase transitions for stochastic block models and reconstruction on trees*, Ann. Probab. **53** (2025), 967–1018. doi:10.1214/24-AOP1723; arXiv:2212.03362.

[9] OpenAI, *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model*, preprint, October 5, 2026. Manuscript, source, and exact certificate programs at commit fd4aeeb2.

[10] OpenAI, *A Capacity Criterion for Four-State Potts Reconstruction on Trees*, preprint, October 5, 2026. Manuscript and source at commit fd4aeeb2.